Processing Pipelines

APPN Aerial SOP — Locked revision 1.0

Processing Pipelines

This page documents the standardised GRYFN processing pipeline YAML files used within APPN for UAV-based data processing. These YAML configurations define consistent, transparent, and repeatable processing workflows across GRYFN-supported sensors and platforms, supporting reproducibility, quality assurance, and cross-project comparability. The files are intended to be used as reference and default templates for approved processing pipelines, with any deviations explicitly documented to maintain traceability and data integrity.

The standard pipelines and the data products they output are detailed below. For a tabular summary of output formats, resolutions and software compatibility, see Standard Data Products.

CALViS — Standard Processing Pipeline

LiDAR-derived products

LiDAR Digital Surface Model (DSM)

- **Output:** `LiDAR_DSM.tif`
- **Type:** Raster
- **Resolution:** 8 cm (fixed resolution)
- **Extent:** VNIR scene extent

LiDAR Digital Terrain Model (DTM)

- **Output:** `LiDAR_DTM.tif`
- **Type:** DTM
- **Resolution:** 100 cm (1 m)
- **Extent:** Processing extent

Combined LiDAR Point Cloud

- **Output:** `LiDAR_CombinedPointCloud.las`
- **Type:** Combined point cloud
- **Notes:** Outliers removed during combination

Hyperspectral products

VNIR Orthomosaic

- **Output:** `VNIR_Orthomosaic.bin`
- **Type:** Orthomosaic (ENVI format, radiance-scaled)
- **Resolution:** 4 cm (GSD-based)
- **Notes:** binning = 2, radiometric calibration applied

SWIR Orthomosaic

- **Output:** `SWIR_Orthomosaic.bin`
- **Type:** Orthomosaic (ENVI format, radiance-scaled)
- **Resolution:** 4 cm (GSD-based)
- **Notes:** binning = 2, radiometric calibration applied

GOBI — Standard Processing Pipeline

LiDAR-derived products

LiDAR Digital Surface Model (DSM)

- **Output:** `LiDAR_DSM.tif`
- **Type:** Raster
- **Resolution:** 8 cm (fixed resolution)
- **Extent:** VNIR scene extent

LiDAR Digital Terrain Model (DTM)

- **Output:** `LiDAR_DTM.tif`
- **Type:** DTM
- **Resolution:** 100 cm (1 m)
- **Extent:** Processing extent

Combined LiDAR Point Cloud

- **Output:** `LiDAR_CombinedPointCloud.las`
- **Type:** Combined point cloud
- **Notes:** Outliers removed during combination

RGB & hyperspectral products

VNIR Orthomosaic

- **Output:** `VNIR_Orthomosaic.bin`
- **Type:** Orthomosaic (ENVI format, radiance-scaled)
- **Resolution:** 4 cm (GSD-based)
- **Notes:** binning = 2, radiometric calibration applied

RGB Orthomosaic

- **Output:** `RGB_Orthomosaic.tif`
- **Type:** Orthomosaic
- **Resolution:** 0.6 cm (fixed resolution)
- **Notes:** Feature-matching (SIFT) bundle adjustment applied